Relatedness networks from international trade data: product space 2018 vs. 2020

Zorka Albert, EC2026 Extra Homework 2

1. Data

OEC/BACI bilateral trade data (HS6, 1996 revision) for 2018 and 2020. I dropped countries with total exports below USD 1 billion (2018 sample: 158 countries, 4,947 HS6 products; 2020 157 countries, 4,763 products), and products with RCA in fewer than 5 countries. This left 4,711 products (nodes) in the 2018 network.

2. Relatedness Network

Computed RCA from co-occurrence as in Hidalgo et al. 2007.

3. Visualizations



Figure 1. *Product space 2018 (HS6, 1996 rev.). Edges: maximum spanning tree, strongest proximity links. Node colour: HS section; node size: global export value in 2018.*

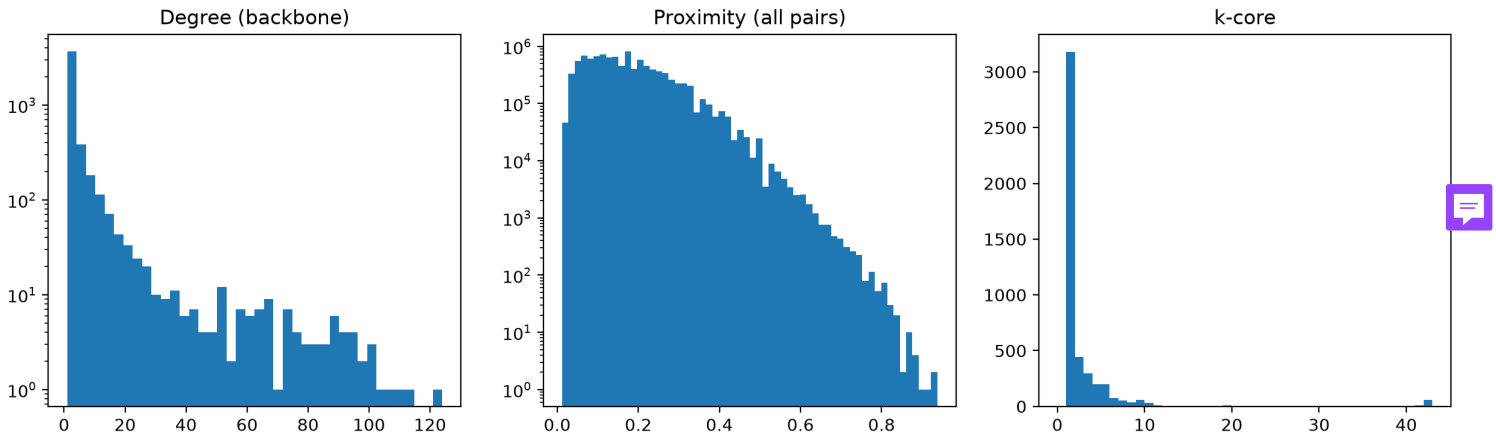


Figure 2. *Left: degree distribution of the backbone (log y-axis). Middle: distribution of all pairwise proximities (log y-axis). Right: k-core decomposition of the backbone.*

The degree distribution is strongly right-skewed, the mean degree is only ≈ 5.1 and the largest hubs reach degree 97-124. The proximities range from 0.01 to 0.93, with millions of weak pairs and an almost exponential decline can be seen at high values. Pairs with $\phi > 0.6$ are rare therefore, strong links are exceptional. Louvain community detection found 77 communities with high modularity (0.77). Among the 6 largest we can find Machinery/electronics, chemicals, textiles and food/ beverages. Machinery is split across three communities.It suggests that products are related through shared production capabilities. The k-core distribution is very uneven, as about 3200 products are in the periphery while a small core of products (59) has k = 42 (all of them belonging to knitted and woven apparel).

5. (Plus) Comparison: 2018 vs. 2020

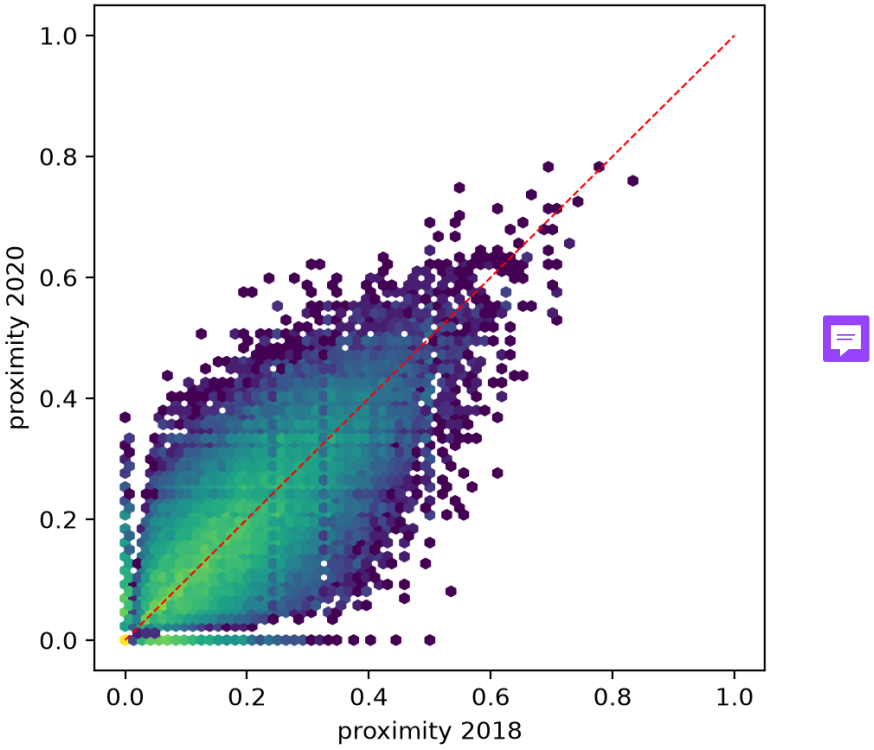


Figure 3. *Proximity of each product pair in 2018 (x) vs. 2020 (y), hexbin density (log colour scale), for products present in both years. Dashed line: y = x.*

Relatedness is a stable structural feature (high correlation), while the order of the top links is noisy. This noise comes from binary RCA, the two-year time horizon, and the 2020 trade shock caused by COVID-19. These cannot be disentangled with just two years.